Concentration Effect in Youth Crime

October 15, 2026

Note: This is not written in a "paper tone", but more as a report to registry the reasoning and the data

1 Research question

Does the concentration of students leaving school at the same time affect crime in the short run?

Reduced form. Did the 2014 change in the schedule of 1st-year EM classes, which stopped them from leaving school at the same time as the 2nd and 3rd years, change crime by or against 2nd- and 3rd-year students after release, in schools that used to release all EM years together?

2 Motivation

School reduces youth crime in two ways: it raises human capital in the long run (Lochner and Moretti, 2004), and it keeps students in a supervised environment, where they can not commit crime (incapacitation). But school also brings young people together, generating networks. Billings et al. (2019) studes the effect of school in formation of network using the redrawing of school boundaries in Charlotte-Mecklenburg in 2002. They find that one standard deviation more same-school, same-grade, same-race and same-gender peers living within 1 km raises the probability of arrest at ages 16–21 by 3.9 percentage points (23%). Two students who live within 1 km of each other are more likely to be arrested together (in the long-run) if they are assigned to the same school.

Concentration can also raise crime in the short run, while students are together. The literature shows evidence of the concentration effects through variation in session-days and the type of crime. Jacob and Lefgren (2003) show that propery crime falls by 14% on school days, but juvenile violent crime rises by 28%, and the violent crime peaks at about 3 p.m., when school lets out. The authors arugue that the fall in property crime is an evidence of incapacitation effects, and the rise of violent crime is an effect of concentration. Luallen (2006) follows the same line and shows that the property crime increases and violent crime decreases on strike days.

I believe this is weak evidence of a concentration effect for some reasons. First, they identify only the net effect of incapacitation and concentration. The effect of school on violent crime is the sum of incapacitation ($-$) and concentration ($+$). Since the net effect is positive, concentration outweighs incapacitation for violent crime. The effect on property crime is the sum of incapacitation ($-$) and concentration, whose sign is unknown. If we assume it is non-negative, the negative net effect means that incapacitation outweighs concentration for property crime. Under these sign assumptions, they identify only the lower bound of the concentration effect.

Second, the source of variation is a whole day without school, which changes substantially what students do during the day, not only how many other students they meet. The changes in property

and violent crime can therefore reflect the different opportunities students face on an "atypical" day: the same student can change which crime they commit. For example, a student who would have fought at school (violent crime) may instead shoplift at a mall (property crime), because a free day leaves time to comute to the mall that is far away. If students substitute across crimes in this way, the type of crime does not identify whether incapacitation or concentration is at work.

Third (and less severe), although Jacob and Lefgren (2003) argue that the bias of school-reporting by estimating the effect of non school hours crimes or crimes that occured out of schools, their effect decrease with their significance.

Separating concentration from incapacitation is hard. Any policy that changes when students leave school usually also changes how long they stay there, and so how much time they spend supervised. This paper uses a source of variation that changes concentration while keeping school and supervision fixed. Two advantages of this paper: incapacitation is held fixed, concentration is measured, not inferred. We could have a dose-response of the concetration of the youth (check that, the first spec would be binary)

Understanding the contemporaneous effect of concentration matters for two reasons:

1. It is cheap to prevent. If shared release raises crime, staggering bell times is a low-cost policy, unlike lengthening the school day or reassigning students across schools.

2. It may feed the long-run effect. If crimes committed together after school are where partnerships start, the short-run concentration effect is one channel behind the networks documented by Billings et al. (2019). Luallen (2006) finds that juveniles who become repeat offenders are more likely to commit their first crime on a strike day.

3 Mechanisms

I can think of three mechanisms on which conecetration of youth can increase crime. First, is the numeric concentration of people, second the peer effect and third the report. Each of these mechanisms operate in the different manner.

A. First, the number of people: how many young people are around. A concentration of people can increase crime because:

   (a) there are more interactions, which can help youths coordinate crimes. Jacob and Lefgren (2003) find no evidence that school increases the coordination of crime: crimes with several offenders do not respond more to school days. We can test whether the number of crimes committed by more than one student increases.

   (b) there are more people who can fight each other. We can test this, but we cannot separate it from old rivalries (the peer effect below).

   (c) there is less adult supervision per youth (police, school staff, security guards on the street), as long as the police do not respond to the crowd.

   (d) more students can attract offenders, which increases crime against students. We can check whether the number of student victims increases.

B. Second, the peer effect: .

   (a) **Learning.** Students can learn from other students. In our main specification, we are measuring the exposition of older students to younger ones so this channel may be weaker in our case. But we can test (long-shot) if there is an increase in crimes committed jointly with a prior offernder and a first-time offender. Perhaps we do not have enough variation.

(b) **Recruitment.** If a 3rd-year student pulls 1st-year students into a theft, the 3rd-year student's own crime also increases, because they offend together. We can test this with crimes committed by mixed-grade groups.

(c) **Status.** Older students may act out/show off to impress younger ones.

(d) **Within student rivalries.** Students can fight because of their rivalries, including hazing of 1st-year students by older students. This predicts fights across grades, with older offenders and younger victims. We can test this.

C. Third, policing and reporting (I am not sure if we treat this as a mechanism or a confounder...)

- **Reporting.** A crowd can change the share of crimes that reach the police, in either direction. It can decrease reporting: offenders are harder to identify in a crowd and each witness expects someone else to call Also, fights between students that would otherwise happen away from the school may happen at the school gate when students are released together. There, the school is more likely to handle them as a discipline matter, so they are less likely to be reported to the police. It can increase reporting: there are more witnesses and more phones to film or call.
- **More police.** When students leave together, the police can respond with more patrols at the school gate. This has two effects: more police can deter crime (a real change in behavior), and more police can detect more,

We can test for these channels by checking whether, with shared release:

(a) the average time between the crime and the arrival of the police (or the report) changes;

(b) the share of crimes in which the police are already at the scene, or arrest the offender in the act (*flagrante*), increases;

(c) the share of incidents started by the police (patrol) rather than by a call from a victim or witness changes;

(d) serious crimes, which are reported whoever is watching , respond in the same way as minor crimes. If only minor crimes respond, the effect is more likely to come from reporting.

4 Further analysis/Heterogenous

We can also investigate:

1. **Displacement.** Does staggered release reduce crime, or only shift it to other times and days? Jacob and Lefgren (2003) and Luallen (2006) find no temporal displacement of violent crime (leads and lags, and weekly and monthly totals). We can run the same tests.

2. **Place.** Where does concentration-driven crime happen: at the school gate, or along the routes between school and home?

3. **Type of crime.** Is the effect driven by fights between students (assaults with young victims) or by property crime? Is it larger for crimes with several juvenile offenders? Jacob and Lefgren (2003) find no evidence that school helps juveniles coordinate crimes. Billings et al. (2019) find that partnerships are common among older youth.

4. Crime against students

5. Powered effect of concentration: if have two shifts in the smae school. The program makes students go earlier in the school in the afternoon shift (starts at _)

5 Population of Study

- **Population.** Students in the 2nd and 3rd years of high school (EM) in state schools in Minas Gerais that offer all three years of EM.
- **Treatment.** The class is released at the same time as the 1st-year EM classes of the same school and shift. A continuous version is the number of 1st-year students released within X minutes of the class.
- **Outcome.** Crime committed by (and against) 2nd- and 3rd-year students after release, near the school and in the minutes after the bell.
- Level of analysis: school

Sample. State schools in Minas Gerais with at least one 1st-, 2nd- or 3rd-year EM class in 2013, excluding the schools of the *Reinventando o Ensino Médio* pilot (2012 and 2013). Table 1 describes the sample. The analysis uses the 1,325 of these schools that, in 2013, had EM classes in the morning and none in the afternoon; they may also have night classes (Table 2).

Table 1: State schools with high school (EM) in Minas Gerais, 2013.

State schools with high school, 2013	N	%
1. All schools	2,209	100.0
2. Pilot schools	132	6.0
3. Schools with afternoon EM classes	551	24.9
4. Schools with different release times within grade	95	4.3
Sample (1 - 2 - 3 - 4)	1,431	64.8

Notes: Schools with at least one 1st-, 2nd- or 3rd-year EM class (regular, integrated or teacher training) in 2013. The sample excludes the schools of the *Reinventando o Ensino Médio* pilot (2012 and 2013). Percentages are relative to the sample. Duration rows: change in the mean duration of 1st-year classes from 2013 to 2014 in at least one shift of the school; unchanged means the same mean duration in every shift with 1st-year classes in both years. Morning: classes starting before 12:00; afternoon: classes starting between 12:00 and 14:00. Release times are compared within the same grade and shift, morning and afternoon only. Average classes per grade: mean over school × grade.

6 Main Identification Strategy

6.1 Motivation for the Identification Strategy

The naive comparison. The simplest design compares schools that release the three EM years together with schools that release each year at a different time. But release times are not random, and this comparison is biased by confounders at school, individual, time of day levels.

*School level.*Schools with fewer classes can release everyone together, so shared release is more common in smaller schools. Smaller schools are in less dense areas with less crime. Staggered times

Table 2: Sample schools by the shift of their EM classes, 2013.

Non-pilot schools by EM shift, 2013	N	%	Also night EM
Non-pilot schools	2,077	100.0	1,344
Only morning	1,325	63.8	823
Only afternoon	146	7.0	66
Morning and afternoon	405	19.5	254
Neither (night only or no time)	201	9.7	201

Notes: Sample of Table 1. Shift of the 1st-, 2nd- and 3rd-year EM classes: morning, classes starting before 12:00; afternoon, classes starting between 12:00 and 14:00; night, classes starting after 14:00. Only morning: at least one morning class and no afternoon class; only afternoon: the reverse. The rows can also have night classes; the last column counts them. Percentages are relative to the sample.

Table 3: Sample schools that release the 1st year together with the 2nd and 3rd years, 2013–2015.

Year	Comparable schools	Release within 5 min	%
2013	1,162	1,151	80.4
2014	1,220	344	24.0
2015	1,315	1,192	83.3

Notes: Sample of Table 1. For each school, year and shift (morning or afternoon), difference between the mean release time of 1st-year classes and the mean release time of 2nd- and 3rd-year classes. Release within 5 min: the difference is 5 minutes or less in at least one shift. Comparable schools: schools with 1st-year and 2nd- or 3rd-year classes in the same shift in that year. Percentages are relative to the sample.

often can come from lack of rooms or extra shifts, which are also related to school quality. Principals may stagger release because of fights at the gate, so treatment responds to past violence (checkif this reverse causality impact the main specification).

- **Size and density.** Schools with fewer classes can release everyone together, so shared release is more common in smaller schools. Smaller schools are in less dense areas with less crime. This biases a comparison of shared and non-shared schools towards finding that concentration *reduces* crime. The bias reverses if treatment is measured as the *number* of students released together, which is larger in large schools located in dense areas.
- **Infrastructure.** Staggered times often come from lack of rooms or extra shifts, which are also related to school quality.
- **Reverse causality.** Principals may stagger release *because* of fights at the gate, so treatment responds to past violence.
- **Nearby schools.** Other schools releasing at the same time add to concentration, and their students may also be affected.

Individual level.

- **Sorting into schools.** Families choose schools, so students in small schools with shared release differ from students in large schools (income, distance from home, prior behavior).
- **Sorting into shifts and classes.** Schools assign students to shifts and classes by age, repetition, behavior or work status. If some classes are released at different times, the students in them may differ in their risk of crime.

Time of day. A different release time is a different clock time, with different parental supervision, daylight, policing and street activity.

A source of variation: the 2014 change in 1st-year schedules. In 2014, 1st-year EM classes in state schools in Minas Gerais became about 50 minutes longer, for that year only (probably the sixth period of *Reinventando o Ensino Médio*; to be confirmed). Morning 1st-year classes moved their release from 11:20–11:30 to about 12:10–12:20, and afternoon classes from 17:00–17:30 to 17:50–18:20. Start times did not change, and most 2nd- and 3rd-year classes kept their schedule, leaving at 11:20–11:30 and 17:00–17:30. In 2015 the 1st year went back to its previous schedule.

For a 2nd- or 3rd-year student in a school that released all EM years together, this means:

- **Concentration falls in 2014 and returns in 2015.** The 1st-year students no longer leave with them in 2014, and leave with them again in 2015.
- **Their own schedule is unchanged.** Start time, class duration and release time are the same, so incapacitation (hours in school, free time) and clock time are held fixed *for the treated students*.
- **The change is set by the state, not by the school.** It does not respond to violence at a particular school's gate.

Design. Compare 2nd- and 3rd-year students in schools that shared release with the 1st year before 2014 (treated: sharing stops in 2014 and returns in 2015) with students in schools that did not share release (control: no change), in 2012–2016. The reversal in 2015 is a second test: if crime falls in 2014 because of lower concentration, it should rise again in 2015.

- **Comparison:** treated vs control schools, before, during and after 2014.
- **Unit:** 2nd- and 3rd-year class (or school × shift) × year.
- **Fixed effects:** school × shift and year.
- **Identifying assumption:** without the 2014 change, crime around the release of 2nd- and 3rd-year students would have evolved in parallel in treated and control schools; and the 2014 change affected these students only through the number of students released with them.

Treated and control schools. The sample is the 1,325 schools with EM classes in the morning and none in the afternoon in 2013 (Table 2). The criteria use only their morning classes, in schools with 1st-, 2nd- and 3rd-year morning classes in 2013 and 2014. Start time, class duration and release time are the means over the morning classes of each grade in the school.

- **(i) Shared release in 2013:** the 1st year left at most 5 minutes before or after the 2nd year and the 3rd year.
- **(ii) The 1st year leaves later in 2014:** the 1st year left 40 minutes or more after the 2nd and the 3rd years, and its class duration rose by 40 minutes or more from 2013 to 2014.
- **(iii) The 2nd and 3rd years do not change:** same start time and same class duration in 2013 and 2014, and so the same release time.
- **Treated:** (i), (ii) and (iii).

- **Control:** (i) and (iii), and in 2014 the 1st year still left at most 5 minutes before or after the 2nd and the 3rd years.

This control differs from the one in the design above (schools that did not share release): here, control schools shared release in 2013 and kept sharing it in 2014. The identifying assumption becomes parallel trends between schools that changed the 1st-year schedule and schools that did not; it fails if the reason a school did not change also changes its crime trend. Table 4 counts the schools that meet each criterion.

Table 4: Treated and control schools: schools with EM in the morning and not in the afternoon, 2013.

Schools	N	%
Sample (Table 1)	1,431	100.0
1st, 2nd and 3rd years in the morning, 2013 and 2014	1,079	75.4
(i) Shared release in 2013	1,069	74.7
(ii) 1st year leaves 40+ min later in 2014	692	48.4
(iii) 2nd and 3rd years unchanged	843	58.9
Treated: (i), (ii) and (iii)	573	40.0
Control: (i) and (iii), still shared in 2014	221	15.4
Other: (i) and (iii), neither	46	3.2

Notes: Only morning: sample schools of Table 1 with 1st-, 2nd- or 3rd-year EM classes starting before 12:00 and none starting between 12:00 and 14:00 in 2013 (Table 2); they may have night classes. The criteria use only morning classes. Percentages are relative to the only-morning sample. Criteria (i)–(iii), treated and control as defined in the text. Other: schools that meet (i) and (iii) but neither (ii) nor the control condition (for example, in 2014 the 1st year leaves between 5 and 40 minutes after the 2nd and 3rd years).

Threats.

- **Treated and control schools differ.** Schools that shared release are smaller (see above). Fixed effects absorb level differences, but not different trends. Check pre-trends in 2010–2013.
- **The 1st-year students also change.** In 2014 they leave later and spend more time in school, so their own crime changes. The outcome must be crime by (or against) 2nd- and 3rd-year students, not all youth crime near the school.
- **Other changes in 2014.** The policy may have come with other changes at the school (activities, staff) that affect all students. In addition, Belo Horizonte hosted World Cup matches in June–July 2014, with changes to the school calendar and heavy policing, and 2014 was an election year. Year fixed effects absorb what is common to all schools, but not effects that differ between treated and control schools.
- **Some 2nd-year classes also change.** The 3rd-year schedule does not change in 2014, but a small share of 2nd-year classes also leaves later that year (Figures 2 and 3). For these classes, the students' own release time changes too.
- **Compliance.** Not all schools changed the 1st-year schedule in 2014 (Tables 1 and 3). Use the observed change in each school's schedule, not just the year.

Descriptive evidence on the 2014 change. Table 1 counts the sample schools whose mean 1st-year class duration rose 50 minutes or more from 2013 to 2014: 1,135 of 2,076 schools (54.7%). The sample excludes the schools of the *Reinventando o Ensino Médio* pilot (2012 and 2013), which may have adopted the change before the rest of the network.

Figure 2 shows that, in the morning, 1st-year classes moved from 11:20–11:30 to 12:15–12:20 in 2014 only, while 2nd- and 3rd-year classes kept leaving at 11:20–11:30. Figure 3 shows the same in the afternoon: 1st-year classes moved from 17:00–17:30 to 17:50–18:20, while 2nd- and 3rd-year classes kept their release time. In the earlier sample of the Belo Horizonte metropolitan region, some schools also moved the afternoon start time earlier in 2014 (from 13:00 to 12:30), for the 2nd year as well as the 1st; this still has to be checked in the current sample.

Figure 1 compares, within each school, the release time of the 1st year with that of the 2nd and 3rd years in the morning shift. In 2013 and 2015 almost every school releases all EM years at the same time (difference of zero). In 2014 most schools release the 1st year about 50 minutes after the 2nd and 3rd years: the median difference is 50 minutes, and 1,067 of 1,654 schools have a difference of 40 minutes or more. 450 schools keep releasing all years together in 2014.

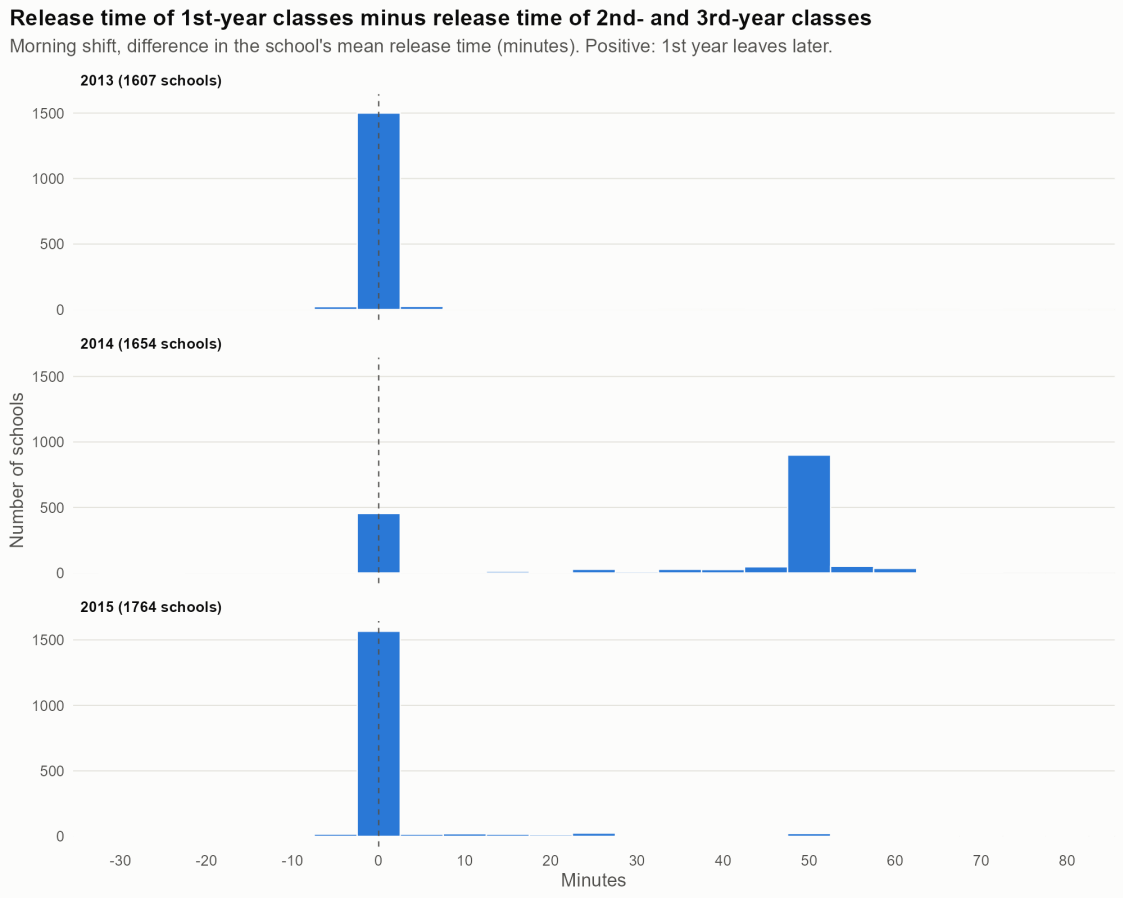


Figure 1: Difference, in each school, between the mean release time of 1st-year EM classes and the mean release time of 2nd- and 3rd-year EM classes, morning shift (classes starting before 12:00), in 2013, 2014 and 2015. Positive values: the 1st year leaves later. Sample schools (Table 1) with both 1st-year and 2nd- or 3rd-year morning classes in the year. The horizontal axis is cut at −30 and 80 minutes, which leaves out 15 schools in 2013, 8 in 2014 and 13 in 2015.

7 Release times by EM grade

7.1 Backup sources of variation

The main design rests on two assumptions: (i) without the 2014 change, crime around the release of 2nd- and 3rd-year students would have evolved in parallel in treated and control schools; (ii) the 2014 change reached these students only through the number of students released with them. Its control group is also small: in 2013, only 79 of the 1,766 comparable sample schools did not release the 1st year within 5 minutes of the 2nd and 3rd years (Table 3). Each design below relaxes part of this. In all of them, V_{jst} is crime by or against 2nd- and 3rd-year students of school j after release in shift s and year t.

(a) Schools that changed vs schools that did not change in 2014.

- **Comparison:** among schools that released the 1st year with the 2nd and 3rd years in 2013, those whose 1st-year schedule changed in 2014 (treated) vs those whose 1st-year schedule did not change (control: 19.6% of the sample have the same 1st-year class duration in 2013 and 2014, Table 1).
- **Unit and fixed effects:** school × shift × year; school × shift and year (regional office × year if the schools that did not change cluster in some regions).
- **Equation:** event study around 2014,

$$V_{jst} = \alpha_{js} + \lambda_t + \sum_{k \neq 2013} \beta_k \, \text{Change}_{js} \times \mathbf{1}[t = k] + \varepsilon_{jst}. \tag{1}$$

- **Identifying assumption:** without the 2014 change, schools that changed and schools that did not would have had parallel crime trends.
- **Violated if** the reason a school did not change its schedule (lack of rooms, a full-time program, a night shift, its regional office) also changes its crime trend. Check pre-trends in 2010–2013 and balance on size, location and region.

(b) Within school, across shifts.

- **Comparison:** in schools with EM in the morning and in the afternoon (19.5% of the sample, Table 1), the shift whose 1st year stopped leaving with the 2nd and 3rd years in 2014 vs the shift where they kept leaving together.
- **Unit and fixed effects:** school × shift × year; school × year, school × shift and shift × year.
- **Equation:**

$$V_{jst} = \gamma_{jt} + \alpha_{js} + \delta_{st} + \beta \, \text{Lost}_{js} \times \mathbf{1}[t = 2014] + \varepsilon_{jst}, \tag{2}$$

  where $\text{Lost}_{js} = 1$ if, in shift s of school j, the 1st year stopped leaving with the 2nd and 3rd years in 2014.
- **Identifying assumption:** within a school, crime after release in the two shifts would have evolved in parallel.
- **What it adds:** school × year fixed effects absorb everything that changed at the school or in its neighborhood in 2014, including other parts of the program that reached all students.

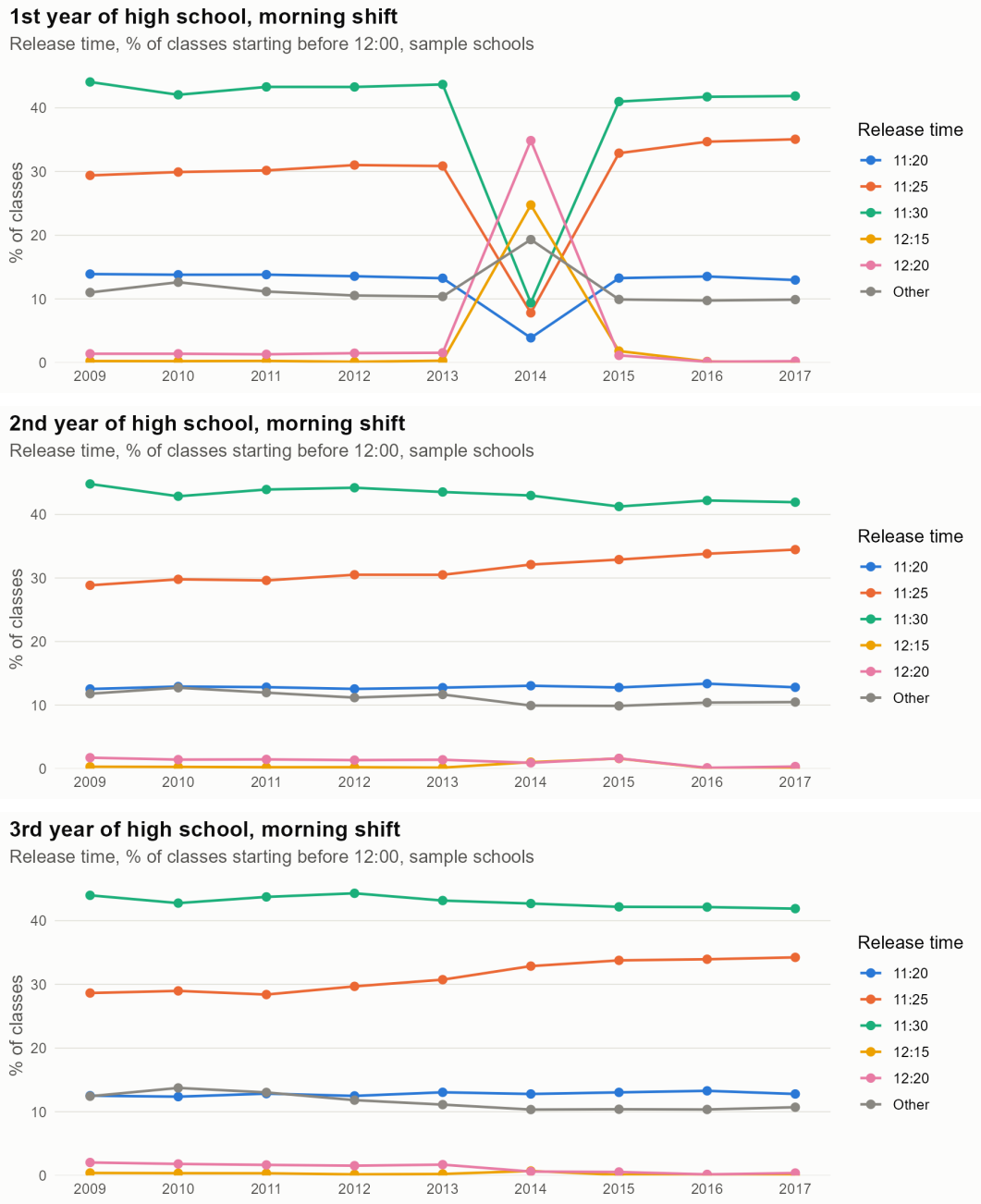


Figure 2: Release time of EM classes in the morning shift (classes starting before 12:00), by grade: 1st year (top), 2nd year (middle) and 3rd year (bottom). Share of the grade's classes released at each time, by year. Sample schools (Table 1). The same release times, colors and vertical axis are used in the three graphs; less frequent times are grouped as Other.

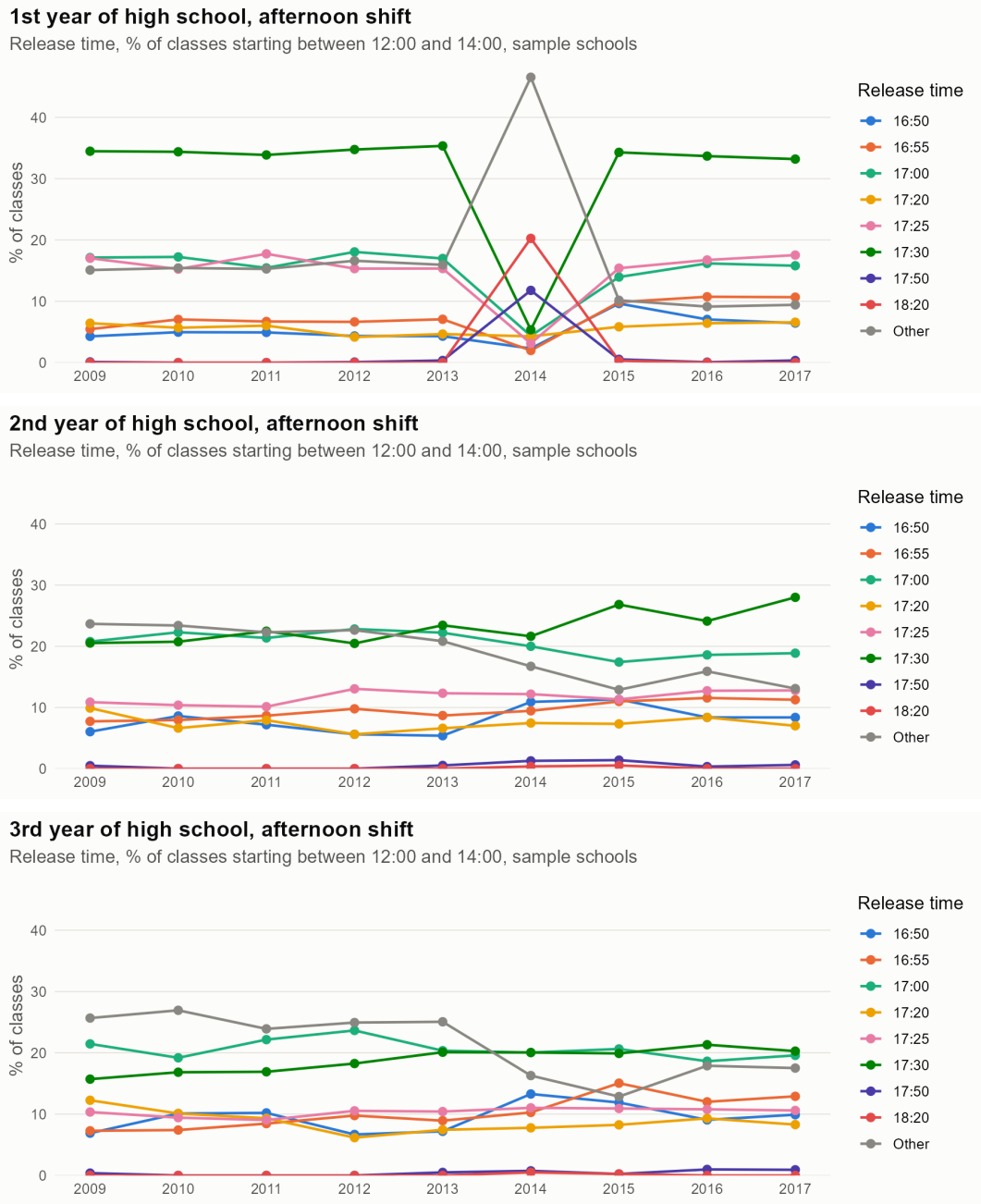


Figure 3: Release time of EM classes in the afternoon shift (classes starting between 12:00 and 14:00), by grade: 1st year (top), 2nd year (middle) and 3rd year (bottom). Share of the grade's classes released at each time, by year. Sample schools (Table 1). The same release times, colors and vertical axis are used in the three graphs; less frequent times are grouped as Other.

- **Violated if** morning and afternoon trends differ for other reasons (who attends each shift, policing by time of day, World Cup match times), or students move between shifts because of the change. Spillover: in 2014 the morning 1st years leave at 12:15–12:20, close to when afternoon students arrive, so the afternoon shift may be affected by the morning change.
- **Feasibility:** count the schools in which the change differs between shifts.

(c) Intensity among schools that changed.

- **Comparison:** only schools that changed in 2014; those that lost a large 1st-year crowd vs those that lost a small one.
- **Treatment:** N_{js}, the number of 1st-year students who left within 5 minutes of the 2nd and 3rd years in shift s of school j in 2013 (measured before the change).
- **Unit and fixed effects:** school × shift × year; school × shift and year.
- **Equation:**

$$V_{jst} = \alpha_{js} + \lambda_t + \beta\, N_{js} \times \mathbf{1}[t = 2014] + \varepsilon_{jst}. \tag{3}$$

- **Identifying assumption:** among schools that changed, crime trends do not depend on the size of the 1st-year crowd they lost (strong parallel trends; Callaway et al., 2024).
- **What it adds:** the comparison does not use schools that did not change, and it gives a dose–response.
- **Violated if** the size of the 1st year tracks school size and density, whose areas have different crime trends; or if a large 1st year relative to the 3rd year signals repetition and dropout, that is, riskier students. Control for 2nd- and 3rd-year enrollment, or scale N_{js} by it.

(d) The 2015 reversal, robust to linear trends.

- **Comparison:** each school's 2014 against the average of its 2013 and 2015, treated vs control schools.
- **Unit and fixed effects:** school × shift × year; school × shift and year.
- **Estimator:**

$$\hat{\beta} = \left[\overline{V}^{T}_{2014} - \tfrac{1}{2}\big(\overline{V}^{T}_{2013} + \overline{V}^{T}_{2015}\big)\right] - \left[\overline{V}^{C}_{2014} - \tfrac{1}{2}\big(\overline{V}^{C}_{2013} + \overline{V}^{C}_{2015}\big)\right]. \tag{4}$$

- **Identifying assumption:** in 2014, the deviation from a linear trend would be the same in treated and control schools. A linear trend that differs between the two groups cancels.
- **Violated if** the effect persists into 2015 (rivalries or partnerships formed in 2014), students anticipate the change in 2013, or a shock specific to 2014 hits the two groups differently.

(e) Time windows around the bell. Needs geocoded crime records with the time of the event.

- **Comparison:** crime near the school in the window after the 2nd- and 3rd-year release vs other windows of the same day, on school days vs days without school (weekends, holidays, vacations), in 2014 vs 2013 and 2015.

- **Unit and fixed effects:** school × 30-minute window × day; school × day, and school × window × type of day (school day or not).
- **Identifying assumption:** a change in 2014 in crime that is specific to the release window *and* to school days comes from the change in who is released.
- **What it adds:** school × day fixed effects absorb anything that hits the school's surroundings on that day; days without school absorb changes in policing or street activity at that hour.
- **Concentration test:** in 2014 the 1st year and the 2nd and 3rd years leave in two windows instead of one. If crime grows faster than the number of students released together, crime in the two windows in 2014 adds up to less than crime in the single window of 2013 and 2015. If crime is proportional to the number of students, the sum does not change.
- **Violated if** police patrols follow the bell and changed in 2014, or the school calendar changed (drop the World Cup weeks of June–July 2014). In the concentration test, the 1st years also leave later and after more hours in school, which changes their own crime.

(f) Student panel. Needs a student identifier across years and crime linked to students.

- **Comparison:** the same students from the 2nd year (2013) to the 3rd year (2014), in schools that changed vs schools that did not.
- **Unit and fixed effects:** student × year; student and year.
- **Identifying assumption:** moving from the 2nd to the 3rd year changes crime in the same way in both groups.
- **What it adds:** student fixed effects absorb the sorting of students into schools.
- **Violated if** dropout, repetition or changes of school depend on the treatment and are related to crime.

Estimators. Treatment switches off in 2014 and back on in 2015, so two-way fixed effects can be biased when effects are heterogeneous; use de Chaisemartin and D'Haultfœuille (2020). If pre-trends fail, re-weight the control schools to match the pre-2014 trends (synthetic difference-in-differences; Arkhangelsky et al., 2021) and report how large a violation of parallel trends would overturn the result (Rambachan and Roth, 2023). If only part of a school's 1st-year classes changed, use the fuzzy version (de Chaisemartin and D'Haultfœuille, 2018).

Which backup for which failure.

- Control schools trend differently: (a), (c), (d) and the estimators above.
- Something else changed at the school or in its neighborhood in 2014: (b) and (e).
- A statewide shock in 2014 (World Cup, elections) that hits treated and control schools differently: (e).
- Sorting of students into schools: (f).

Dropped from the 9th-grade version. The within-school comparison of classes released at different times (the earlier "RCT") has little variation: only 6.2% of the sample schools have different release times within a grade and shift (Table 1). Changes in shared release in years other than 2014 appear to be rare (almost every school releases all EM years together in 2013 and 2015, Figure 1), and they are chosen by the school, which brings back reverse causality.

References

Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager, "Synthetic Difference-in-Differences," *American Economic Review*, 2021, *111* (12), 4088–4118.

Billings, Stephen B., David J. Deming, and Stephen L. Ross, "Partners in Crime," *American Economic Journal: Applied Economics*, 2019, *11* (1), 126–150.

Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna, "Difference-in-Differences with a Continuous Treatment," NBER Working Paper 32117, National Bureau of Economic Research 2024.

de Chaisemartin, Clément and Xavier D'Haultfœuille, "Fuzzy Differences-in-Differences," *Review of Economic Studies*, 2018, *85* (2), 999–1028.

_ and _, "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects," *American Economic Review*, 2020, *110* (9), 2964–2996.

Jacob, Brian A. and Lars Lefgren, "Are Idle Hands the Devil's Workshop? Incapacitation, Concentration, and Juvenile Crime," *American Economic Review*, 2003, *93* (5), 1560–1577.

Lochner, Lance and Enrico Moretti, "The Effect of Education on Crime: Evidence from Prison Inmates, Arrests, and Self-Reports," *American Economic Review*, 2004, *94* (1), 155–189.

Luallen, Jeremy, "School's Out...Forever: A Study of Juvenile Crime, At-Risk Youths and Teacher Strikes," *Journal of Urban Economics*, 2006, *59* (1), 75–103.

Rambachan, Ashesh and Jonathan Roth, "A More Credible Approach to Parallel Trends," *Review of Economic Studies*, 2023, *90* (5), 2555–2591.